Land Use and Land Cover Mapping in China Using Multimodal Fine-Grained Dual Network

Shang Liu, Huadong Wang, Yuan Hu, Mengting Zhang, Yixuan Zhu, Zhibin Wang, Dongyang Li, Mingyang Yang, and Fan Wang

***Abstract*— With the advancement of geo-systems and the increased availability of satellite data, a plethora of land-use and land-cover (LULC) products have been developed. The existing LULC products primarily relied on time-series imagery to classify land by pixel-based classifiers, allowing for local analysis and accurate boundary detection. However, the advent of deep learning has shifted toward the use of patch-based CNN models for generating land cover maps. In this article: 1) we create a training dataset for China using a voting strategy based on three off-the-shelf available LULC products, avoiding the labor-intensive manual annotation. 2) We design a novel CNN-based model for the LULC task, called multimodal fine-grained dual network (dubbed as Dual-Net), which takes dual-date images to generate final maps and reduces the need for gap-free temporal sequences or separate cloud detection. To leverage the correlation between location, date, and category, we embed modal information (dates and geo-locations) into the model. Furthermore, by incorporating low-level constraints and using pseudolabel refinement, we continually improve the performance and achieve more refined segmentation. 3) Due to the lack of a suitable validation dataset for China, we create a new validation dataset called the China Sentinel2 Validation Dataset (CSVD) by manually annotating 733 finely labeled images of 1024 × 1024 pixels of China-specific Sentinel2 data. 4) Extensive experiments demonstrate that our model outperforms existing LULC products and produces more fine-grained segmentation results comparable to other patch-based products. Finally, we release annual LULC maps for China in 2020–2022 and also make our model accessible online for real-time results export.**

***Index Terms*— Attention mechanism, dual net, land-use and land-cover (LULC), multimodal information fusion, pseudolabel refinement, refined dataset construction.**

I. Introduction

A CRUCIAL challenge for Earth observation is determining land-use and land-cover (LULC), which is essential for a variety of fields, including resource management [1], sustainable development [2], ecosystems [3], climate change [4], and food security [5]. In addition, monitoring changes and trends in LULC can offer valuable information for government, businesses, and farmers' decision-making.

Manuscript received 10 February 2023; revised 8 May 2023; accepted 31 May 2023. Date of publication 14 June 2023; date of current version 6 July 2023. *(Shang Liu and Huadong Wang contributed equally to this work.) (Corresponding author: Shang Liu.)*

Shang Liu, Yuan Hu, Mengting Zhang, Yixuan Zhu, Zhibin Wang, Dongyang Li, Mingyang Yang, and Fan Wang are with the DAMO Academy, Alibaba Group, Beijing 100102, China (e-mail: liushang.ls@alibaba-inc.com; lavender.hy@alibaba-inc.com; zhangmengting.zmt@alibaba-inc.com; zhuyixuan.zyx@alibaba-inc.com; zhibin.waz@alibaba-inc.com; yingtian.ldy@alibaba-inc.com; david.ymy@alibaba-inc.com; fan.w@alibaba-inc.com).

Huadong Wang is with the Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: huadw2012@163.com).

Digital Object Identifier 10.1109/TGRS.2023.3285912

As satellite technology has advanced, it has become easier to access various satellite data. Some examples of coarse data include VEGETATION (1-km resolution) from SPOT-4 [6], [7], MODIS (500-m resolution), medium resolution imaging spectrometer (MERIS) full resolution (300-m resolution) [8], [9], and PROBA-V (100-m resolution) [10], [11]. More recently, moderate resolution data, such as Landsat (30-m resolution) [12], [13] and Sentinel2 (10-m resolution) [14], [15], have become readily available.

This has led to the creation of a variety of LULC products [16], with resolutions ranging from coarse to high. Some examples of these products include global land cover (GLC)-2000 [17], [18], Geocover-2000 [19], GlobCover-2009 [20], GlobeLand30 [21], FROM-GLC [22], [23], GLC-FCS30 [24], and the newly released 10-m resolution products of FROM10 [25], the European Space Agency (ESA)'s world cover [26], Environmental Systems Research Institute (ESRI) [27], and dynamic world [28].

In addition to publicly released LULC maps, several LULC benchmark datasets have also been published. EuroSAT [29] presents a European dataset based on Sentinel-2 satellite imagery. The Gaofen Image dataset (GID) [30] provides 150 Gaofen-2 satellite images with a 0.8-m resolution. OpenEarthMap [31] introduces a global high-resolution LULC mapping dataset containing 5000 images at 0.25–0.5-m resolution. Kaiser et al. [32] leverages OpenStreetMap (OSM) to automatically derive weakly labeled training data from Google Maps for three LULC classes. iSAID [33] offers the first benchmark dataset for instance segmentation in aerial images. The MillionAID [34] dataset is a large-scale aerial image dataset created by researchers at Carnegie Mellon University, Pittsburgh, PA, USA, containing over 1 million high-resolution aerial images from various data sources. These benchmarks provide valuable resources for research in LULC classification and related fields.

Earlier LULC classification products were generated primarily using pixel-based techniques. Numerous studies have explored various pixel-based machine-learning methods for this purpose. Early on, the parametric maximum likelihood classifier (MLC) [35] was used to estimate class probabilities for each pixel. To improve results, nonparametric classifiers were later employed, including artificial neural networks (ANNs) [36], decision trees (DT) [37], random forests (RF) [38], and support vector machines (SVMs) [39]. These data-driven classifiers learned complex decision rules from training data to generate class maps. In recent years, deep learning models, such as convolutional neural networks

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

(CNNs) [40], have gained popularity for land cover mapping. In contrast to pixel-based methods, CNNs receive an image patch as input and estimate the class of the entire patch, which is referred to as a patch-based approach. Several studies have applied CNNs for land cover classification. For instance, Li et al. [41] utilized CNNs to integrate Google Earth imagery with Landsat data to improve 30-m resolution land cover mapping. Verma and Jana [42] employed a simple CNN to map complex urban forms. ESRI [27] made use of UNet [43] to generate global LULC maps, while Dynamic World [28] trained a fully convolutional network (FCN) [44] to produce near real-time LULC predictions. Moreover, L2HNet [45] implemented a low-to-high network for high-resolution land-cover mapping using low-resolution labels.

When comparing the two approaches, the data format and classification principle between the pixel-based and patch-based approaches differ significantly. Pixel-based training data are constructed by sparse sampling, and when predicting, this approach processes each pixel independently and identifies the class of pixel by constructing time series information. This approach can provide more detailed information and can identify tiny objects, but it is more sensitive to noise in the training data and has less ability to generalize. While patch-based methods employ annotations of the entire image as training targets and employ cropped image patches for testing, they capture context information and show a better ability to generalize over space and time. Furthermore, this approach typically runs inference on a single image, making it easy to implement and capable of producing predictions in near real time. However, deep learning models often suffer from a loss of detail as the network becomes deeper, resulting in a larger mapping unit and lower spatial accuracy.

The deep learning methods rely on data from patch-based visual interpretation, which is labor-intensive and time consuming. Labeling medium-resolution satellite data, which lacks texture and subtle details, is particularly challenging and requires specialist knowledge and additional support from high-resolution imagery such as Google Maps or Bing Maps. If even finer annotations are needed, more human resources must be invested. For example, Dynamic World used 25 experts and 45 commissioned labelers to produce 24 000 training tiles and 409 validation tiles. However, there are only 1085 training tiles and 13 validation tiles in China, which is insufficient for training and testing. Landcovernet [46] uses a consensus algorithm to produce 2875 annotated tiles in Asia using a multisource satellite image as a reference, but in China, there are only 739 tiles of size 256×256, which is not enough for training.

As a result of the limited availability of annotated data in China, existing data are not suitable for training and testing deep learning models. To address this issue, we propose using existing LULC products to generate training samples. We select two pixel-based products (From10, ESA-2020) and one patch-based product (ESRI) as candidates and use them to construct patch-based training data through a voting strategy. This strategy allows us to filter out pixels with semantic conflicts and obtain high-quality training data.

Rather than use a single image like ESRI and Dynamic World, we design a dual network, which flexibly handles one or two satellite images as input. This model can achieve superior performance and reduce the influence of clouds, and it also can produce results in real time. We also incorporate a multimodal information fusion block to incorporate date and location information and employ a low-level attention (LLA) refinement to ensure that the deep feature distribution is similar to the source image. To address the issue of coarse segmentation in deep learning models, we implement pseudolabeling and spectral indexing to refine the labels and improve the model's performance. These techniques allow us to continuously improve the model's performance and produce refined segmentation results.

To validate our product, we develop a carefully labeled validation dataset[1] and employ two public training datasets from LandCoverNet and Dynamic World. and achieve 69.59, 72.29, and 68.74 mIoU on each of these datasets, outperforming other products. Finally, we release a new LULC product[2] in China from 2020 to 2022. We also provide an online interface for researchers to access results in real-time.[3]

In summary, our main contributions include the following.

1) We offer a solution for LULC classification that strives to achieve a low cost while maintaining relatively higher accuracy, and which can be adapted to other regions of the world.
2) We develop a novel deep learning model, called multimodal fine-grained dual network (Dual-Net model), that uses two images from different dates and apply multimodal information fusion and low-level attention blocks to further improve performance.
3) We provide an annotated validation dataset containing 733 finely labeled patches, which is currently the largest of its kind in China and helps to address the lack of validation data in the country. We released annual land cover maps for China in 2020–2022 and also provide an online interface for users to generate real-time maps.
4) Extensive experimentation and visual comparisons have shown that our Dual-Net framework performs better on multiple validation datasets and produces finer segmentation than other patched-based LULC products.

II. Materials

A. Study Area

China's geography [47] is incredibly diverse, with a range of different types of land cover, including plains in the north, snow-capped mountains and high plateaus in the west, and terraced hills and river valleys in the south. The country's climate is also varied, with desert-like conditions in the north and west, and tropical monsoon weather in the south and southeast. This complex geography and climate make it difficult to create accurate LULC maps in China. Besides, the rapid pace of development and urbanization in the country adds another layer of complexity to the task of mapping

[1] https://engine-aiearth.aliyun.com/research/dataset_csvd.
[2] https://engine-aiearth.aliyun.com/#/dataset/DAMO_AIE_CHINA_LC.
[3] https://engine-aiearth.aliyun.com/#/portal/analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

LULC. Despite these challenges, creating accurate and up-to-date maps is essential for a variety of applications, including environmental protection, natural resource management, and disaster risk reduction.

B. Training Dataset Construction

In recent years, with the free availability of remote sensing imagery, such as Landsat and Sentinel2, various global land-cover maps have been produced, including GlobeLand30, FROM-GLC30, GLC-FCS30, ESA, FROM10, and ESRI. These products differ in their classification taxonomy, input features, classifiers, or overall spatial agreement. Despite these efforts, it remains challenging to generate high-quality LULC maps on an annual basis. To create refined products, we have selected three high-resolution products with a similar first-level class classification scheme. Two pixel-based products, ESA-2020 and FROM10, are chosen for their ability to identify fine features, such as rivers and roadways. One patch-based product ESRI applies CNN to generate annual maps by compositing predictions from multiple dates, achieving an overall accuracy of 85%. By combining these products according to a voting policy, we can eliminate inconsistent predictions with conflicting labels and preserve consistent pixels as training examples. This method allows us to achieve a better balance between local details and overall accuracy.

All three of the land-cover classification systems we have selected using a classification scheme consisting of ten first-level classes defined according to the United Nations Land Cover Classification System (LCCS) [48]. The FROM10 classification system is similar to the ESA taxonomy, with the key difference being the class *tundra*. Since *tundra* only exists in a few areas of China, such as Changbai Mountain, we ignore it from our class schema. Compared with ESRI, there are two major differences in the classification of certain land cover types. First, *rice fields* and *irrigated/inundated farms* are belong to the category of *flooded vegetation* in ESRI, but are classified as *cropland* by ESA and FROM10. Second, *grasslands* in ESA and FROM10 must have herbaceous cover percentages above a certain threshold (15% for FROM10 and 10% for ESA), and land below this threshold is classified as *bareland*. In ESRI, however, a mix of small clusters of plants or solitary plants scattered throughout a landscape with exposed rock or soil may also be referred to as *rangeland* (grassland).

Additionally, *shrub-scrub* and *wetland* often lack distinctive appearances and are easily confused with other types, leading to a limited agreement among the different classification systems. Table I summarizes the accuracy of *shrubland* and *wetland* classes, which show lower performance. After voting, there are just a few viable labels left. Therefore, we have removed *shrub-scrub* and *flooded vegetation* (wetland) from our classification schema. Our final classification taxonomy is similar to those of FROM10 and ESA, including *crops*, *trees*, *grassland*, *water*, *building* (artificial surfaces), *bareland*, and *permanent snow/ice*.

To capture the diverse land types of China, we uniformly split the country's map into a 3 × 3 degree grid and randomly select 18 locations as anchors. These anchors are slightly adjusted to cover rare classes (such as *snow*) and to avoid the entire patch being covered by a single category. Fig. 1 shows the resulting sample grid. At each sample location, we have collected Sentinel-2 images from 2020 with image cloudiness of less than 20%. Due to the high cloudiness in Tibet in 2020, we have used images from 2019 instead. Finally, we have created our training dataset by cropping a Sentinel-2 GeoTiff raster with a 1024 × 1024 image patch.

TABLE I
PERFORMANCES OF THE CLASSES *Shrubland* AND *Wetland* IN THREE LULC PRODUCTS. IN THE ESRI PRODUCT, THE CLASS *Flooded Veg* IS USED INSTEAD OF *Wetland*

Dataset	Class	User Accuracy	Producer Accuracy
ESA	Shrubland	44.10	38.60
	Wetland	40.60	27.80
FROM10	Shrubland	62.75	50.77
	Wetland	34.48	10.14
ESRI	Scrubland	83.11	48.5
	Flooded Veg	57.56	73.21

III. METHODOLOGY

Deep learning models, particularly CNNs, have achieved impressive results in a variety of computer vision tasks. When applied to the field of remote sensing, these models [45], [49], [50], [51] have also demonstrated remarkable breakthroughs. For example, the multi-granularity multi-level feature ensemble network (MGML) [49] utilizes multilevel features to achieve the highest accuracy in scene classification tasks. The Factseg [50] uses an activation-driven approach to identify small objects in remote sensing segmentation. The L2HNet [45] has successfully created 1-m resolution land-cover maps using 30-m resolution land-cover products as training labels. Given the generalizability of current deep learning models, we have also explored their performance on land cover tasks.

In this section, we propose a deep learning framework (Dual-Net) for the LULC classification task, which consists of two parts.

1) **Multimodal fine-grained dual network (Dual-Net model)**, a deep neural network model that takes two images as input, consisting of three parts: *CNN-based feature extraction backbone*, *low-level constraint*, and *multimodal information fusion*.
2) **Pseudolabel refinement**, considering the lack of fine-grained object annotations (such as roads, rivers, and so on) in the training data, pseudolabels are introduced from OSM to assist the Dual-Net model in training and improve the model's prediction performance.

A. Dual-Net Model

LULC is highly dynamic due to changes in natural phenology and human activities, with the landscape continually evolving over time. Land cover classes, such as *snow* and *water*, are easily affected by seasonal fluctuations, resulting in dynamic changes in their coverage area. Moreover,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

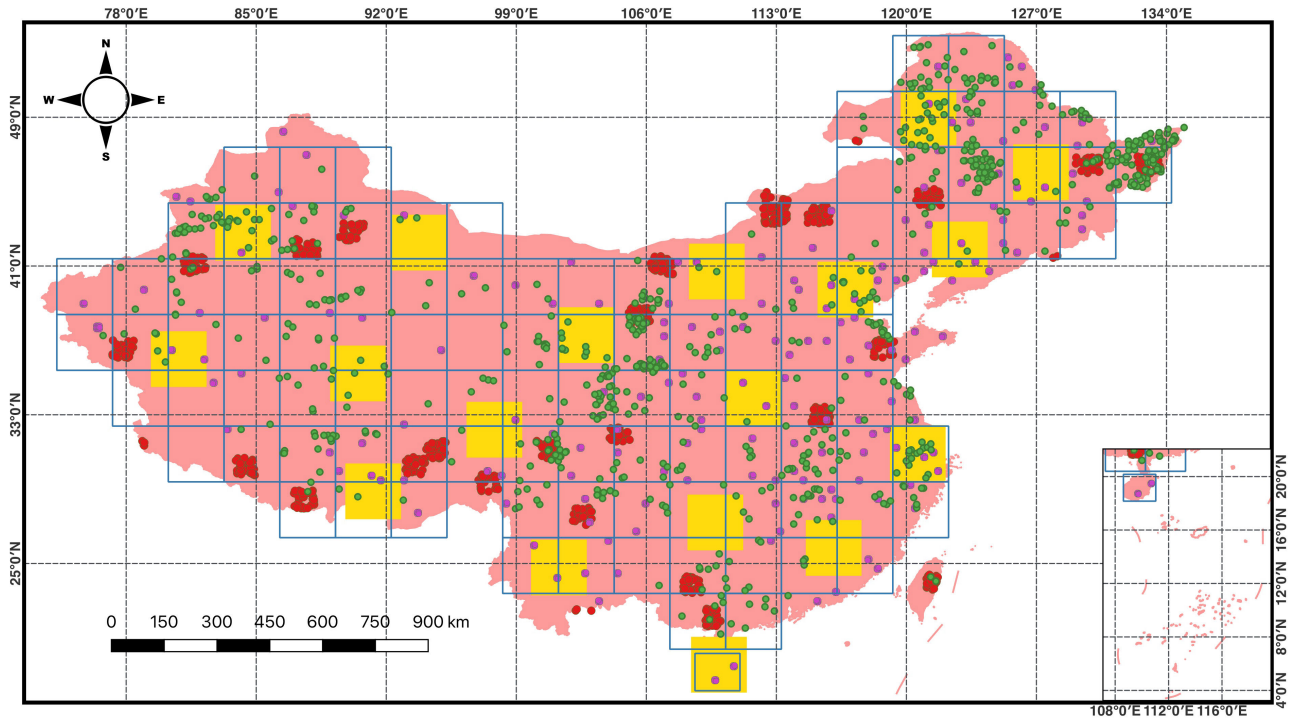


Fig. 1. Distributions of training and validation data. Blue boxes represent uniformly split grids with size 3 × 3 degrees. Yellow patches represent 18 training sites. Red dots represent 739 validation tiles from 27 sites of LandCoverNet. Purple squares represent 733 validation tiles from 208 sites of CSVD. Green dots represent 768 validation tiles from 768 sites of Dynamic World.

vegetation (e.g., *grass* and *trees*) is sensitive to date and location, displaying different phenological cycles among various vegetation types compared with nonvegetation classes. Additionally, land use classes are typically constant and not affected by these changes. For example, a rice field in the grain-filling, growing, and harvesting periods should consistently be classified as crops, rather than being classified as *water*, *crops*, or *bareland* in each respective period. To leverage the temporal characteristics inherent in LULC data and reduce the influence of human activities (such as sowing and harvest), major pixel-based methods identify classes by using a time series of images. However, this approach is time-consuming and requires processing numerous images. In contrast, our low-cost, data-friendly Dual-Net model captures the temporal characteristics of LULC data using only a pair of images from different dates, without requiring any data preprocessing steps.

The architecture of the Dual-Net model is shown in Fig. 2(a). First, we create a CNN-based model using high resolution network (HRNET)-small [53] as the backbone. HRNET builds a high-resolution branch in its initial stage and gradually extends it with a low-resolution branch in subsequent stages, allowing it to generate multiresolution features. To capture more details, we add an extra higher-resolution branch from the initial stem convolution layer of HRNET, which allows the highest output resolution to reach half of the input resolution. Next, the Dual-Net uses two parallel backbones to extract the multiresolution features of the two image inputs separately. Third, we concatenate features at the same level from two backbones and feed it to a BIPFN [52] module. BiFPN forms bidirectional cross-scale connections between feature maps of different resolutions, addressing the absence of explicit cross-scale connections in conventional CNNs. It incorporates both an upsampling path and a downsampling path. Given the multiscale features of the dual-net backbones, we initially concatenate features of the same scale. Following this, we apply a 3 × 3 convolution layer to reduce the channel dimensions of the concatenated features. Subsequently, we implement the bottom-up upsampling path, which upsamples lower resolution features from the top of the backbone to match the size of higher resolution features. We then execute the top-down downsampling path, performing the reverse operation. Ultimately, the upsampled and downsampled feature maps are combined using element-wise addition. By facilitating the flow of information up and down the feature pyramid, BiFPN allows features to acquire richer contextual and spatial information from other scales. This bidirectional exchange of information culminates in a more robust feature representation that benefits various visual understanding tasks. Finally, we append an upsample layer at the end of BIFPN and apply two convolution layers to produce the final predictions.

Since date and location have a substantial influence on land cover types, we insert a multimodal information fusion block into different HRNET stages to incorporate date and location information. To further leverage the low-level attribute of remote imagery, we also introduce a multiscale LLA block to guide the generation of deep features.

B. Multimodal Information Fusion

Different regions often have varying climates, which can significantly alter the distribution of land covers. As a result, the date and location of an image have a significant impact

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

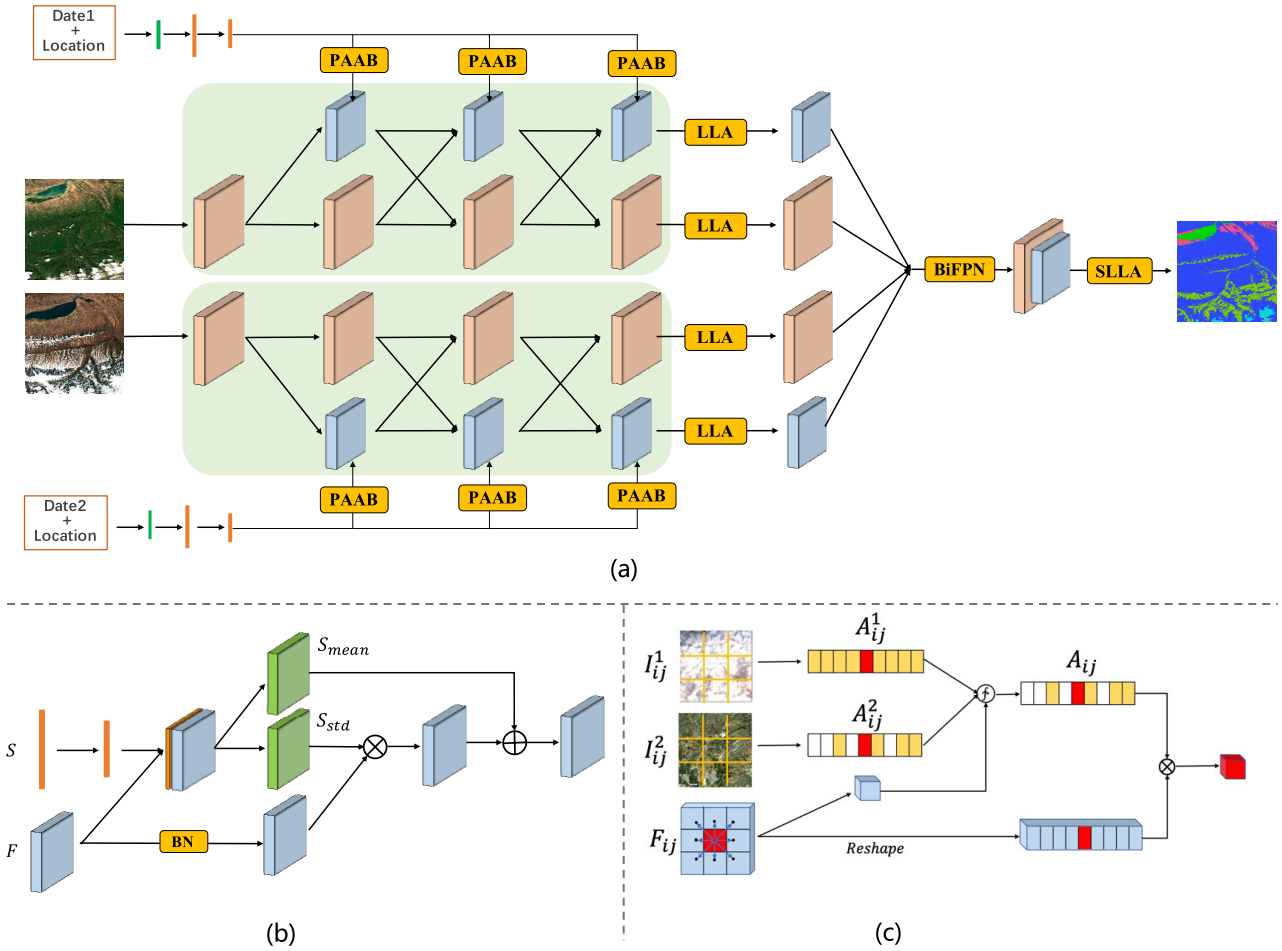


Fig. 2. Architecture of proposed Dual-Net model. The top part (a) shows the Dual-Net architecture, its main components include: CNN-based feature extraction backbone (green regions), multimodal information fusion module (date and location encoding and PAAB), low-level constraint module (LLA, SLLA, and BiFPN [52]). The bottom parts (b) and (c) show the specific structures of the PAAB and SLLA, respectively.

on the distribution of land cover types. Besides, different vegetation at the same location often has similar characteristics, which makes classification difficult. For instance, in the summer, tall lush crops and trees may have similar vegetation indices and visual textures. However, if date information is included in the classification, vegetation categories, such as trees, meadows, and cultivated land, can be easily identified. The combination of multimodal information, such as dates and locations, can improve the accuracy of land cover classification. Many existing deep learning models developed for LULC classification ignore this information. To the best of our knowledge, our model is the first to integrate these multimodal information for the LULC task.

1) Date and Location Information Encoding: To encode the date and location information, we first obtain the date date of the input image, then map it to the range of [0, 1], and next apply three transform functions (linear, sin, and cos) to the initial value. This results in a date code with four elements $[v_1, v_2, v_3, v_4]$

$$v_1 = \text{date}/365$$
$$v_2 = 2(v_1 - 0.5)$$
$$v_3 = \sin((v_1 - 0.5)\pi)$$
$$v_4 = \cos((v_1 - 0.5)\pi) - 0.5.$$

For the location code, we represent the location using longitude and latitude pairs based on the WGS84 Coordinate System (EPSG CODE: 4326). We normalize the location information by setting the range of the bounding box in China to [70, 0, 140, 60]. Unlike the date code, the location code is simply encoded using a linear function

$$v_5 = \text{longitude}/180$$
$$v_6 = (\text{longitude} - 70)/(140 - 70)$$
$$v_7 = \text{latitude}/90$$
$$v_8 = (\text{latitude} - 0)/(60 - 0).$$

For instance, consider an image patch cropped at location coordinates (116.0431° and 39.8023°) on the date May 20, 2021. This date is first converted to the number of days since the start of the year (139), which is then transformed into the date code [0.8676, −0.3576, 0.3835, 0.2328] using v_1, v_2, v_3, v_4. The location code for the latitude and longitude (116.0431° and 39.8023°) is obtained by transforming these coordinates using v_5, v_6, v_7, v_8, resulting in [0.6577, 0.5960, 0.6446, 0.6633].

One might initially consider representing date and location information using only two variables each, resulting in a total

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

of four variables. However, we have opted for eight variables in order to more accurately capture the intricacies of the data. For date encoding, we employ both cosine and sine encoding to represent the cyclical nature of seasons, as this approach allows us to differentiate between similar months (e.g., December and January) and distinguish distinct seasons (e.g., spring and autumn). As for location encoding, we utilize two different amplitude ranges—global (longitude ranges from 0° to 180°, while latitude ranges from 0° to 90°) and local (specific to the Chinese region)—to emphasize geographical differences on both large and small scales.

2) Position-Aware Adaptive Block: We apply a modified adaptive instance normalization module (AdaIn) [54] to incorporate date and location information (i.e., latitude and longitude) into the Dual-Net model. The AdaIn module is typically used for conditional image generation, which takes style features and convolution features as inputs and performs style transfer in the feature space by transferring convolution feature statistics. Given a convolution feature F and a style feature S, AdaIn is defined as

$$\overline{F} = \frac{F - \mu_c(F)}{\delta_c(F)} \tag{1}$$

$$F^s = \gamma_c^s \times \overline{F} + \beta_c^s. \tag{2}$$

Here, $F \in R^{C\times H\times W}$ represents the convolution feature, where C, H, and W denote the channel, height, and width of the feature, respectively. $\gamma_c^s = \text{Linear}(S)$ and $\beta_c^s = \text{Linear}(S) \in R^C$ are two style feature vectors that are generated by linear mapping to replace the affine parameters in the original instance normalization (IN) block. $\mu_c(F), \delta_c(F) \in R^C$ are the channelwise mean and standard deviation, computed across the batch size and spatial dimensions independently for every feature channel.

Instead of using channelwise denormalization used in AdaIn [54] for the generation task, we propose a position-aware denormalization block for the LULC classification task, which is shown in Fig. 2(b). Given style feature S, we repeat S spatially to match the size of the convolution feature F, then concatenate S and F, and send it to two separate convolution layers to generate affine features. One generates position-aware scale features γ_p^s of size $N \times C \times H \times W$, and the other generates position-aware bias features β_p^s. In addition, we replace the original instance normalization with batch normalization, which normalizes all images across the batch and is more friendly for land cover classification tasks. Finally, we use the position-aware scale features γ_p^s and bias features β_p^s to affine the normalized features, as follows:

$$F^s = \gamma_p^s \times \widetilde{F} + \beta_p^s. \tag{3}$$

By this method, we apply position-aware denormalization and make position with different semantics to capture date and location information independently. In short, position-aware adaptive block (PAAB) uses a style-transferring approach to fuse multimodal information by combining deep features and encoded information with position-aware features.

C. Low-Level Constraint

Remote sensing images differ from natural images in that they often have strong low-level characteristics, whereas similar colors and textures typically belong to the same semantic category. However, as the depth of the CNN model increases, detailed information is lost, and the semantic distribution no longer matches the pixel distribution [55]. Attention mechanism is a deep learning technique, which is widely used in natural language processing and computer vision due to its excellent performance. It has achieved better results in many tasks, such as image classification and segmentation than CNN-based models [56], [57]. The attention mechanism can allow the model to selectively attend to certain parts of the input, which can improve performance on tasks that require the model to identify specific objects or features within an image. They work by aggregating features based on an attention map and can be useful in tasks where the model needs to align different parts of the input data in order to make a prediction. Therefore, we design the low-level constraint module based on the attention mechanism to enforce deep features to have the same texture distribution as the original image, which is composed of three main components: LLA, BiFPN [52], and SLLA. This can help preserve important information and improve the performance of the model.

1) Low-Level Attention: Assuming the convolution feature is $F^k \in R^{C\times H\times W}$, where C, H, and W denote the channel, height, and width of the feature, respectively, and $k = 1, 2$ denote the two image inputs. We first resize image I^k to the same size as feature F^k, and then extract a pixel feature that is similar to local binary patterns (LBPs) [58] by calculating the distance between the central anchor pixel I_{ij}^k and nearby neighbor pixels $\{I_{\text{ijm}}^k\}$

$$D_{\text{ijm}}^k = \left|\left| I_{ij}^k - I_{\text{ijm}}^k \right|\right|^2 \tag{4}$$

where $m \in \delta_{ij}$ [the set of nine nearby neighbors of the anchor pixel (i, j)]. Second, we just consider the nearby neighbors with a smaller distance

$$S_{\text{ijm}}^k = \begin{cases} 1 - \dfrac{D_{\text{ijm}}^k}{\sum_{m=1}^M D_{\text{ijm}}^k}, & \text{if } D_{\text{ijm}}^k \le \lambda \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where λ is a distance threshold, and it is set to the average value $\sum_{m=1}^M D_{\text{ijm}}^k / M$ in our experiments ($M = 9$), and S_{ijm}^k represent the similarity score between the anchor and neighbor pixel. Third, we normalize S_{ij}^k as attention map A_{ij}^k and use it to aggregate neighbor features

$$A_{\text{ijm}}^k = \frac{S_{\text{ijm}}^k}{\sum_{m=1}^M S_{\text{ijm}}^k} \tag{6}$$

$$F_{ij}^k = \sum_{m=1}^{M} A_{\text{ijm}}^k F_{\text{ijm}}^k. \tag{7}$$

This process is displayed in Fig. 2(c), where the colorful grid vector represents attention map A_{ij}^k, the red cell denotes the anchor pixel, yellow cells denote similar pixels, and white cells represent pixels that have a large distance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

Because LBP-like features use neighboring pixels, they can only capture short-range relationships. To incorporate long-range relationships, we exploit the hierarchical structure of deep networks to implement LLA at different feature scales, which is called multiscale LLA. Since our backbones generate multiresolution features with $1/2, 1/4, 1/8$, and $1/16$ from the initial input resolution, we downsample the image to the corresponding scale and apply LLA at this resolution.

2) Switch Low-Level Attention: After conducting LLA at two backbone branches, the multiscale features are fed to BiFPN module [52] and produce one final feature with $1/1$ resolution. To address this, we add a dynamic switch to the end of BIFPN, which dynamically selects the appropriate LLA from A_{ij}^k [as shown in Fig. 2(c)]. The output of SLLA is the weighted value of A_{ij}^1 and A_{ij}^2

$$A_{ij} = \alpha A_{ij}^1 + (1-\alpha) A_{ij}^2 \tag{8}$$

where $A_{ij}^k, k = 1, 2$ are generated LLA by two input images, and the weight $\alpha = (\mathcal{F}(F_{ij}))$, where $\mathcal{F}$ is a two-layer MLP.

Finally, the selected attention map is used to update F_{ij}. This process aggregates features with similar pixel values, ensuring that the constrained feature has the same distribution as the image.

D. Pseudolabel Refinement

The current LULC products often neglect minor roads and rivers because annotating tiny objects is tedious and time-consuming. We use *OSM* road and waterway data to refine the segmentation of tiny objects. However, there are many missing annotations in *OSM*, making it impossible to use the data directly. To overcome this issue, we propose using pseudolabels to identify tiny objects. Instead of relying on high-confidence predictions to fill in unlabeled areas, we propose using spectral indexes to merge labels. Spectral indexes have been shown to improve land cover performance, such as using the normalized difference vegetation index (NDVI) for vegetation separation [59] and the normalized difference water index (NDWI) for water detection [60]. The whole workflow of data preparation is shown in Fig. 3.

First, we utilize LULC maps from From10, ESA, and ESRI to employ a label voting strategy, which is inspired by the expert majority voting approach used for validation data acquisition in the Dynamic World. Our voting policy follows a similar principle: at least two candidates from ESA, ESRI, and FROM10 must agree on a label, and the agreed-upon class is then selected as the final training label. Then, we incorporate data from roadways and waterways into the *voted label* to generate *merged label* and use it to train the initial model. Next, we feed train patches to the initial model to generate *predicted label*. Finally, we merge the *merged label*, *ESA label*, and *predicted label* via NDVI to produce the final *pseudolabel*. In detail, we initially calculate the average NDVI of two images during training and consider pixels with NDVI greater than 0.2 and marked as nonvegetation categories as mislabeled areas by merged label. We then replace these pixels with vegetation labels from the *predicted label*. If the associated *predicted label* is still nonvegetation, we choose labels from the *ESA label*. If both the *predicted label* and *ESA label* are nonvegetation, the pixel is marked as 255 and ignored during training. Similarly, pixels with NDVI lower than -0.05 and labeled as vegetation categories, are also regarded as wrong labels, and we use the same previous policy to correct wrong labels. Finally, the updated labels are used to retrain the model. In this way, the final trained model can accurately capture more tiny objects.

IV. Validation Datasets and Evaluation Metrics

A. LandCoverNet Dataset [46]

LandCoverNet[4] is a global dataset for land cover classification from Sentinel-1, Sentinel-2, and Landsat-8 in 2018. The class schema is based on a hierarchical schema created by a group of experts and includes seven classes (*water*, *natural bare ground*, *artificial bare ground*, *woody vegetation*, *cultivated vegetation*, *(semi) natural vegetation*, and *permanent snow/ice*). In Asia, there are a total of 2753 image chips of 256×256 pixels. However, only 739 chips from 27 sites are available in China, as shown as red dots in Fig. 1. LCND is used as a validation set because it matches our classification schema and uses patch-based annotation. We filter out error labels by choosing only high-confidence labels with consensus scores above 71.

LandCoverNet officially provided the labeling results and Sentinel-2 images in 2018. To collect 739 pairs of data with a three-month gap, we select images with the least cloud cover from the months of April, May, August, and September, term the sample data as LCND. Although we chose data from four months, there is still a significant amount of data covered by clouds.

B. Dynamic World Dataset [28]

The Dynamic World[5] is a collection of over five billion human-labeled pixels from Sentinel-2 satellite images in 2019. They used 25 experts and 45 nonexpert annotators to dense markup the images using a ten-class classification schema. The labeling was done using the Labelbox platform, where annotators marked the boundaries of objects directly over the Sentinel-2 tile of 510×510 pixels. In addition to the Sentinel-2 tile, annotators also refer to a matching high-resolution satellite image via Google Maps or Google Street View. The experts produced approximately 4000 image tiles, and the additional 45 annotators labeled 20 000 image tiles. A verification set of 409 image pairs was selected and relabeled by three experts, but it only contains 13 tiles in China and is too small to provide a solid evaluation. Therefore, we use the labeled pairs in China from the training set, with a date range of May–October, to generate 768 annotations, denoted as *Dynamic World Dataset (DWD)*.

C. China Sentinel2 Validation Dataset

Since the LandCoverNet dataset and DWD do not have enough annotations in China and they skip some tiny

[4]https://mlhub.earth/data/ref_landcovernet_as_v1.

[5]https://doi.pangaea.de/10.1594/PANGAEA.933475.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

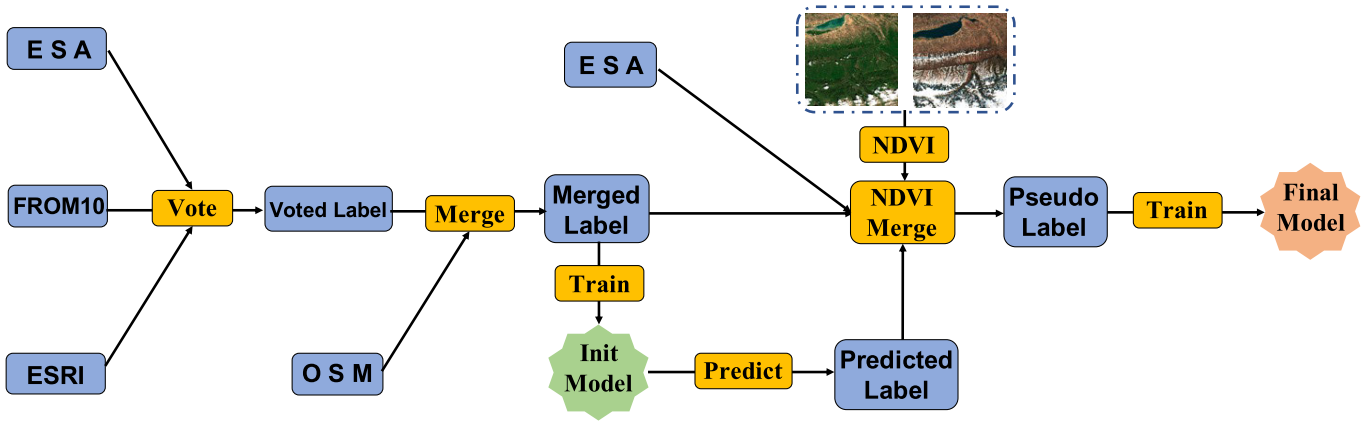


Fig. 3. Workflow of pseudolabel refinement, including five steps: 1) label voting: initial annotation results (called *voted label*) are obtained using a label voting strategy that combines the results from ESRI, FROM10, and ESA. 2) Merging refined labels: The voted Label data is merged with road and waterway tags from OSM to produce refined annotation results called the *merged label*. 3) Initial model training and label prediction: The Dual-Net is trained on the *merged label* data and makes predictions on the training data to obtain predicted labels. 4) Label merging based on the NDVI index (pseudolabeling): The merged label, predicted label, and ESA label data are merged based on the NDVI index to create the pseudolabels. 5) Training the Dual-Net based on the pseudolabel to obtain the final model.

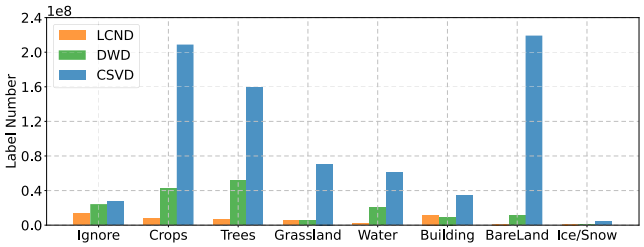


Fig. 4. Label distributions of three validation datasets.

objects, they are not suitable for evaluating refined segmentation. Therefore, we created a finely labeled validation dataset in China called China Sentinel2 Validation Dataset (CSVD), which has a class taxonomy of seven categories (see Section II-B). We randomly sampled 208 sites in China and then cropped four image patches of 1024 × 1024 pixels at each site. We used Google Earth Engine (GEE) [61] to collect the Sentinel2 data for the sampling sites during the months of April, May, September, and October 2021.

Due to cloud interference and missing data, we obtained 733 effective matching pairs of 1024 × 1024 pixels, which we refer to as CSVD. We hired eight annotators with experience in remote sensing annotation and two experts as instructors. When labeling, annotators referenced BGR, SWIR, and NIR images from two periods and were required to annotate small objects, such as narrow roads and riverways. Following the rule of effective ground cover, annotators ignored weather changes, such as clouds and snow. For semantic conflicts caused by seasons or artificial activities, like changes in permanently snow-covered areas and new buildings, we marked them as *ignore*. We carried out three rounds of labeling, with the first two rounds labeled independently and the third round focused on locations where the labels from the previous two rounds were inconsistent.

The category distributions of the three datasets are shown in Fig. 4. Their distributions are similar, with an unbalanced problem where the classes *crops* and *trees* make up the majority of the labeled data and *ice/snow* makes up the least. Although LCND and DWD are from the training set, the CSVD dataset has a much higher quantity of effective annotations. Due to the small image size of 256 × 256 pixels, LCND has the fewest effective annotations. A visual comparison of the three datasets in Fig. 5 shows that even though tiny objects, rivers, and narrow roads are marked, the LCND and DWD datasets ignore these details.

D. Evaluation Metrics

To evaluate and analyze the performance of the proposed method on the LULC tasks, we adopt three commonly used evaluation metrics [21], [22], [62], [63].

1) **Producer accuracy**, also known as **Recall**, is a measure of the ability of a model to correctly identify a certain class of land cover. It is calculated as the number of true positive instances (i.e., instances that are correctly classified as the target class) divided by the total number of instances of that class in the ground truth.
2) **User accuracy**, also known as **Precision**, is a measure of the ability of a model to correctly classify an instance as a certain class. It is calculated as the number of true positive instances (i.e., instances that are correctly classified as the target class) divided by the total number of instances predicted to be that class.
3) **Mean Intersection over Union (mIoU)**, also known as Jaccard Index, is commonly used to evaluate the performance of semantic segmentation tasks. The calculation of IoU is the intersection area between the predicted segmentation and ground truth divided by the union area between them. For multiple classes, the mean IoU is calculated by taking the average of the IoUs corresponding to each class.

In LULC classification, both producer accuracy and user accuracy are useful metrics for understanding the tradeoff between false negative instances and false positive instances.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

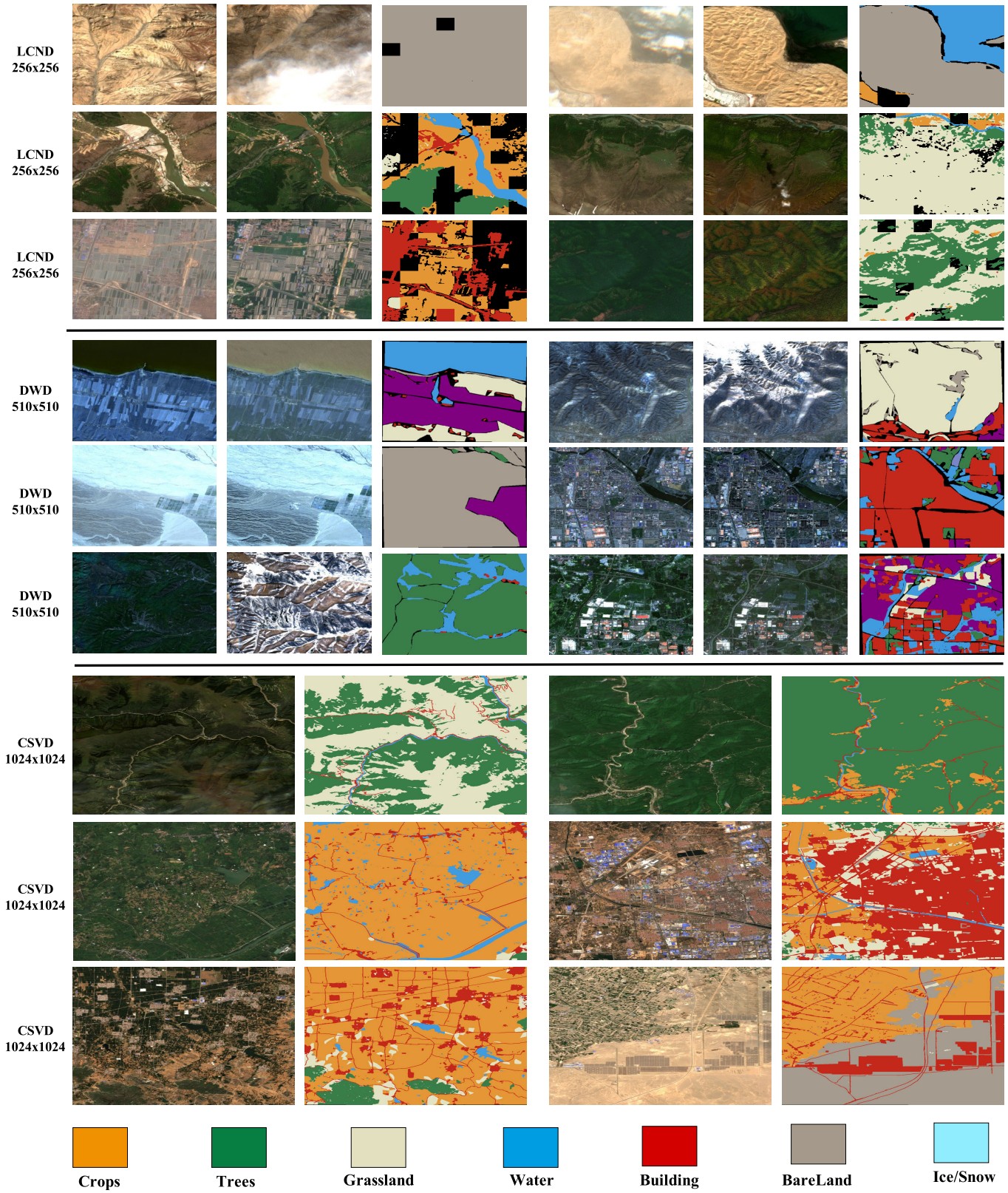


Fig. 5. Visualization of the three validation datasets. The top three rows show the LCND, with an original size of 256 × 256 pixels. The middle three rows show the DWD, with an original size of 510 × 510 pixels. The bottom three rows show the CSVD, with an original size of 1024 × 1024 pixels.

In imbalanced classification tasks (such as segmentation), mIoU is more representative than per-pixel accuracy. Since our task involves image segmentation in an imbalanced label distribution (as shown in Fig. 4) and we evaluate using patch-based annotations, the mIoU is taken to evaluate model performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

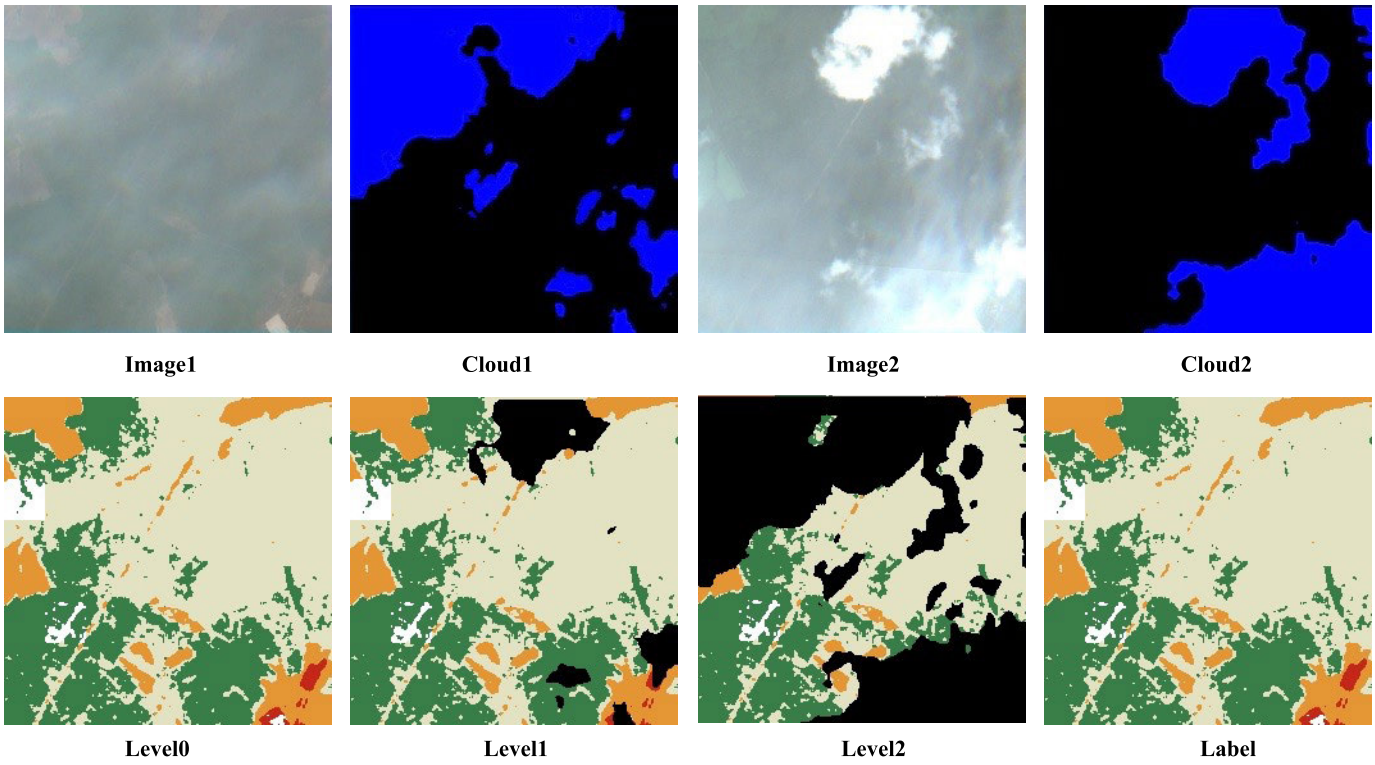


Fig. 6. Illustration of different levels of cloud cover. Image1 and Image2 are a pair of input images, and the corresponding cloud prediction images are shown in Cloud1 and Cloud2, where the blue part represents the cloud cover region. Level0, Level1, and Level2 represent the images with cloud areas removed at three levels of cloud cover, respectively, used for result evaluation.

We follow the approach of ESRI and exclude the class *snow/ice* when evaluating the performance because snow cover can vary greatly depending on the season and weather. As seen in Fig. 4, the label *ice/snow* has a very small proportion in three validation datasets.

V. Experimental Details

A. Image Preprocessing

The Sentinel2 (S2) satellites, launched by ESA, have a revisit time of five days and produce imagery with 13 spectral bands at spatial resolutions ranging from 10 to 60 m. In this study, we select eight optical bands from the Sentinel-2 Level-2A product: Blue (B2), Green (B3), Red (B4), Vegetation Red Edge1 (B5), Vegetation Red Edge3 (B7), NIR (B8), SWIR1 (B11), and SWIR2 (B12). To reduce the influence of snow in winter, we remove winter imagery from the northeast regions (Heilongjiang, Jilin, and Liaoning) and some western regions (Qinghai, Xinjiang, and Tibet). The original Level-2A Sentinel-2 images contain both temporal and spatial metadata, including acquisition date and location. To prepare the training data, we first gathered all Sentinel-2 images within a specified location. These images were then cropped into 1024 $\times$ 1024 pixel patches, with each patch retaining its associated date and geographic information. During training, two image patches acquired on different dates from the same location were randomly selected, and the date and location information were encoded as additional inputs into the dual-neural network. Before feeding to the model, we clip the first four bands to range 0–3000 and the last four bands to range 0–5000, then utilize 3000 and 5000 as normalizers to map the appropriate bands to range $[0, 1]$. Last, we apply normalization to normalize image values, where the mean and standard deviation for the normalized values are 0.500 and 0.3125, respectively.

TABLE II

Effect of Varying Levels of Cloud Cover on the Performance of the Model. "Cloud Ratio" Refers to the Proportion of Images With Cloud Coverage

Datasets	Cloud Level	Cloud Ratio	mIoU	Recall	Precision
LCND	0	0.00	67.97	78.94	82.04
	1	11.61	71.68	82.42	83.44
	2	49.73	**75.53**	**85.76**	**85.25**
CSVD	0	0.00	68.30	80.31	81.84
	1	0.45	68.67	80.64	82.02
	2	6.89	**68.85**	**80.82**	**82.31**

B. Training Details

To optimize our model, we use the AdamW optimizer with a poly learning rate scaling strategy and an initial learning rate of 0.01, momentum of 0.9, and weight decay rate of 5e-4. We set the input size to 768 $\times$ 768, the batch size to 32, and run 40 000 iterations. During training, we use augmentation methods such as random resize with interpolation rates between 0.5 and 1.5, random crop with size 768 $\times$ 768, and gridshuffle [64], which grid number is set to 4. The experiments were carried out on a GPU server equipped with eight NVIDIA Tesla V100 32-GB GPUs. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

TABLE III

EXPERIMENTAL RESULTS OF MODEL ABLATION STUDY. ADAIN REFERS TO THE USE OF THE ORIGINAL ADAIN [54] MODULE FOR MULTIMODAL INFORMATION FUSION. THE CHECK MARK "✓" INDICATES THAT THE CORRESPONDING MODULE HAS BEEN ADDED TO DUAL-NET. IN THE FIRST COLUMN, WE USE THE NUMBERS TO REPRESENT THE MODELS WITH PARTICULAR SETTINGS. THE VALUE IN EACH BRACKET IS THE INCREASED VALUES WHEN COMPARED WITH THE BASIC MODEL 2

No.	Model	AdaIN	PAAB	LLA	PLR	mIoU	Recall	Precision
1	Single-Net					66.49	78.74	79.04
2	Dual-Net					68.67	80.64	82.02
3		✓				67.83 (−0.84)	79.58 (−1.06)	81.09 (−0.93)
4			✓			69.24 (+0.57)	81.61 (+0.97)	82.37 (+0.35)
5			✓	✓		69.53 (+0.86)	81.71 (+1.07)	82.53 (+0.51)
6			✓	✓	✓	**69.59 (+0.92)**	**81.76 (+1.12)**	**82.59 (+0.57)**

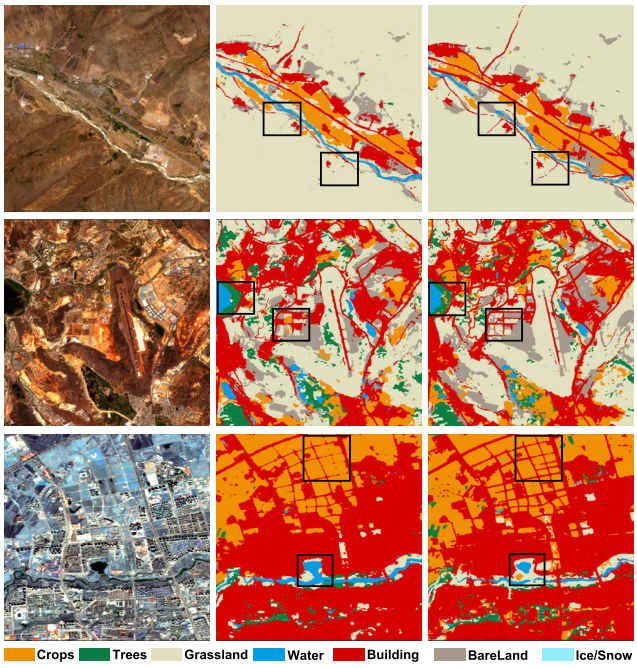


Fig. 7. Visual changes when applying the PLR. The left column shows images from Sentinel2, the middle column shows the visualization without PLR, and the right column shows the visualization with PLR. The significant differences are marked with black boxes.

model was developed using the PyTorch framework and the MMsegmentation library [65] for semantic segmentation tasks.

VI. EXPERIMENTAL RESULTS

A. Effect of Cloud Cover

To evaluate the impact of cloud cover on model performance, we conducted an experiment on different levels of cloud cover. First, we used the self-developed cloud prediction model to predict the cloud distribution on images in the validation set. As shown in Fig. 6, given a pair of input images Image1 and Image2, the corresponding cloud prediction results are Cloud1 and Cloud2 (the blue parts represent cloud distribution). We defined three levels of cloud cover processing based on the cloud distribution of the two images.

1) **Level 0**: Ignoring the cloud cover in images and directly inputting the image into the model to obtain the prediction result, and then evaluate the entire predicted image.
2) **Level 1**: Intersecting (AND) the prediction results of the cloud regions of both images, with any common cloud regions in the two images only being considered as cloud regions. In Fig. 6, Label1 shows the cloud distribution after taking the intersection of Cloud1 and Cloud2, where the nonblack part is used for result evaluation.
3) **Level 2**: Take the union (OR) of the cloud region prediction results of the two images, and any image with clouds is considered a cloud region. In Fig. 6, Label2 shows the cloud region distribution after taking the union of Cloud1 and Cloud2, where the noncloud region part is used for result evaluation.

The proportion of image data (cloud ratio) and the baseline model (Dual-Net) without any previous enhanced modules, evaluation performance under different levels of cloud cover are shown in Table II, from which we can draw the following three conclusions.

1) Our annotated validation dataset CSVD has less cloud cover. The cloud cover images of CSVD at Level 1 and Level 2 only account for 0.45% and 6.59%, respectively. However, the cloud cover images of LCND at Level 1 and Level 2 account for 11.61% and 49.73%, respectively
2) On the LCND, more cloud cover has a greater impact on the algorithm, and the mIoU changes significantly for different cloud cover levels. However, on the CSVD, where there is little cloud coverage, performance does not change significantly.
3) Our model Dual-Net is highly robust to clouds. The results of the model on LCND and CSVD without ignoring any cloud (LCND Level 0) are similar to those of other LULC products (ESA2020 68.43 mIoU and ESA2021 68.14 mIoU, as shown in Table V).

To ensure fair comparisons with other models, we evaluate our follow-up experiments based on Level 1 cloud coverage, which ignores common cloud regions in two input images.

B. Ablation Study

We perform an ablation study to evaluate the contributions of each module in our model. Single-Net represents the CNN-based model with only the backbone part (HRNET-Small) of Dual-Net and taking a single-date image as input, while Dual-Net refers to the proposed model framework

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

COMPARISON OF THE MODEL'S PREDICTION USING A SINGLE-DATE IMAGE AND A PAIR OF IMAGES WITH TWO DATES AS INPUT. "AVG." INDICATES THE AVERAGE PERFORMANCE ACROSS ALL CLASSES

Datasets	Class	One-Date			Two-Dates		
		mIoU	Recall	Precision	mIoU	Recall	Precision
LCND	Crops	68.96	72.73	**93.01**	**78.19**	**83.58**	92.38
	Trees	66.38	85.31	74.95	**73.34**	**89.14**	**80.54**
	Grassland	41.86	66.00	53.36	**46.17**	**69.03**	**58.23**
	Water	**85.20**	**91.14**	92.89	84.95	89.67	**94.17**
	Building	61.78	75.05	77.74	**67.88**	**76.71**	**85.50**
	BareLand	78.95	87.25	89.25	**83.20**	**88.03**	**93.81**
	Avg.	67.19	79.58	80.20	**72.29**	**82.69**	**84.11**
CSVD	Crops	72.52	83.92	**84.22**	**75.38**	**90.41**	81.93
	Trees	78.47	91.63	84.52	**83.17**	**92.13**	**89.53**
	Grassland	51.02	**62.12**	74.07	**51.84**	61.26	**77.12**
	Water	80.65	**91.44**	87.23	**83.20**	89.86	**91.82**
	Building	64.52	**72.32**	85.68	**65.52**	70.03	**91.04**
	BareLand	56.97	83.67	**64.10**	**58.43**	**86.85**	**64.10**
	Avg.	67.36	80.85	79.97	**69.59**	**81.76**	**82.59**

TABLE V

MIOU PERFORMANCE OF LULC PRODUCTS ON THREE VALIDATION SETS. "AVG." INDICATES THE AVERAGE PERFORMANCE ACROSS ALL CLASSES. THE AVAILABLE 10-M RESOLUTION PRODUCTS FROM RECENT YEARS WERE COLLECTED FROM OFFICIAL WEBSITES, INCLUDING FROM10-17 [25], ESRI-20 [27], ESRI-21 [27], ESA-20 [66], ESA-21 [26], AND OURS-21, WITH SUFFIX NUMBERS INDICATING THE YEAR (E.G., FROM10-17 REFERS TO FROM10'S ANNUAL COVERAGE IN 2017). THE LAST COLUMN, LABELED "OURS" DENOTE THE PROPOSED METHOD. SINCE THE DYNAMIC WORLD'S OFFICIAL WEBSITE DOES NOT PROVIDE ALIGNED IMAGES WITH DIFFERENT DATES, THE RESULTS OF OUR METHOD ON DWD ARE NOT PROVIDED HERE. THE BEST RESULTS ARE MARKED IN BOLD

Datasets	**Class**	**From10-17**	**ESRI-20**	**ESRI-21**	**ESA-20**	**ESA-21**	**Ours-21**	**Ours**
LCND	Crops	66.66	65.69	65.15	70.59	70.98	72.94	**78.19**
	Trees	71.52	72.85	71.92	71.05	68.01	**73.91**	73.34
	Grassland	39.53	30.3	30.23	44.77	43.99	**49.44**	46.17
	Water	77.54	79.03	77.85	82.45	80.07	84.81	**84.95**
	Building	52.93	50.5	50.45	62.06	62.97	64.03	**67.88**
	BareLand	79.22	44.83	45.44	79.68	82.84	**87.08**	83.20
	Avg.	64.57	57.20	56.84	68.43	68.14	**72.04**	**72.29**
CSVD	Crops	58.34	66.8	67.24	69.5	70.31	75.01	**75.38**
	Trees	72.13	72.10	70.35	76.59	75.14	82.02	**83.17**
	Grassland	30.13	45.19	44.08	45.96	46.57	50.15	**51.84**
	Water	72.76	75.86	77.87	65.33	76.02	80.02	**83.20**
	Building	41.35	51.50	51.94	47.32	51.34	64.72	**65.52**
	BareLand	46.76	45.36	45.41	44.11	56.75	57.72	**58.43**
	Avg.	53.58	59.47	59.48	58.14	62.69	**68.27**	**69.59**
DWD	Crops	57.67	**75.86**	75.73	70.23	70.87	72.47	-
	Trees	77.97	78.16	77.74	82.65	82.09	**83.21**	-
	Grassland	16.84	21.92	21.45	28.29	25.74	**29.74**	-
	Water	78.53	88.31	87.59	78.69	81.83	**82.65**	-
	Building	41.62	73.88	73.09	50.52	52.75	**65.89**	-
	BareLand	71.22	67.51	66.54	59.96	74.33	**78.46**	-
	Avg.	57.31	67.61	67.03	61.72	64.60	**68.74**	-

(Fig. 2) that uses a pair of two-date images as input. We construct variations of Dual-Net by adding modules on top of the backbone model and train them on the training set, observing their prediction performance on the validation dataset. The components of interest here include the multimodal information fusion module AdaIN/PAAB, the low-level constraint module LLA, and the pseudolabel refinement (PLR).

The experimental results are summarized in Table III, from which we can draw the following conclusions.

1) Dual-Net using two-date images as input significantly outperforms single-net using a single-date image, indicating that considering different date images helps to improve the model's classification ability.
2) Comparing the results of model 3 and model 4 with model 2, using AdaIn decreases the model's prediction performance while using the proposed PAAB improves it by 0.57 mIoU. This suggests that using PAAB as a fusion of multimodal information is better than AdaIn, and that position-aware fusion helps with land cover classification.
3) The results of model 2 and models 4–6 show that the model's prediction performance is improved by gradually adding the PAAB, LLA, and PLR modules, with an increase in mIoU of 0.57, 0.86, and 0.92 respectively. The combination of these modules also achieves the best results, indicating that the introduction of PAAB, LLA,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

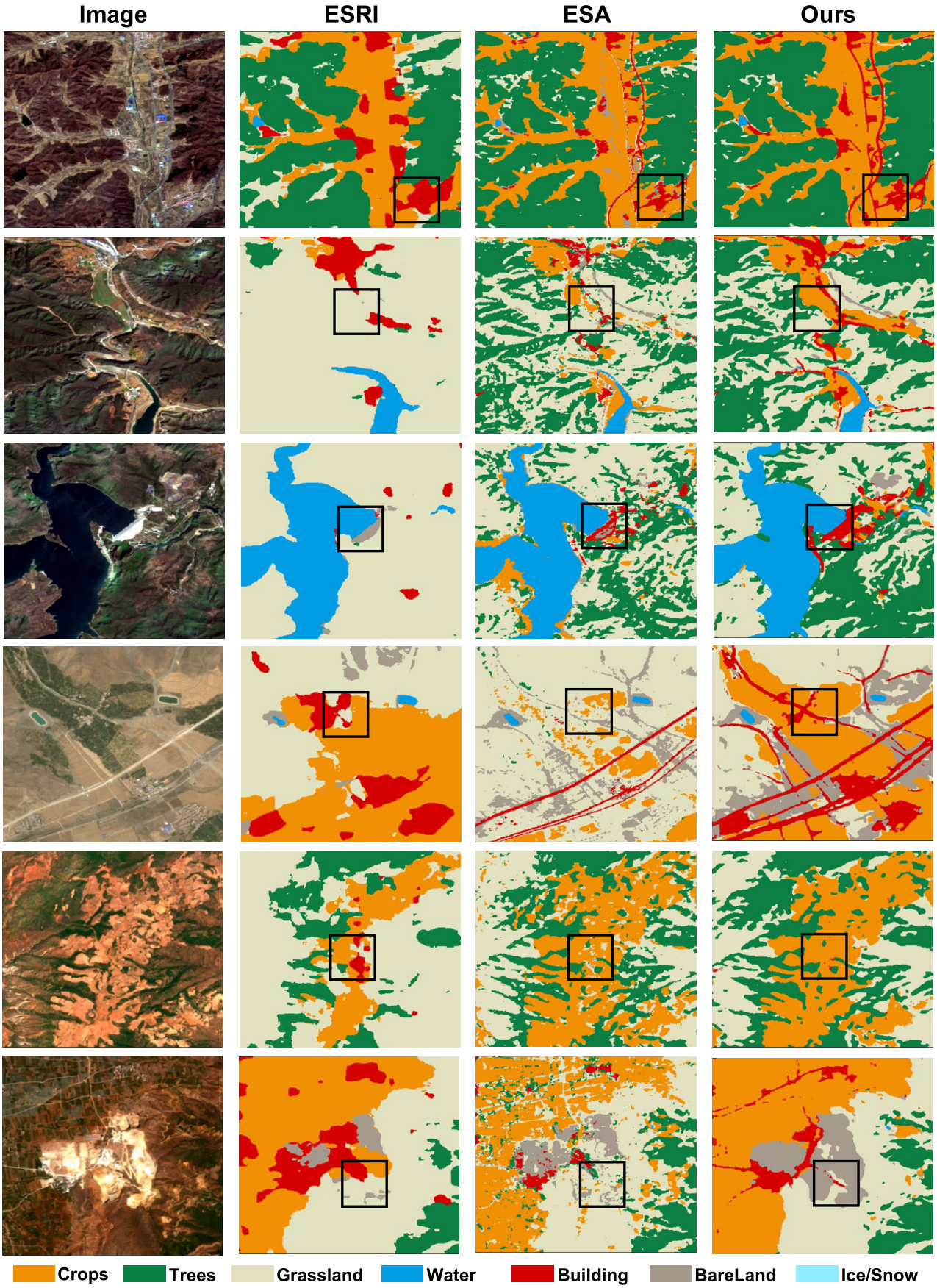


Fig. 8. Local visual comparison of the three products. The first column is Sentinel2 images. The second to fourth columns display the results of ESRI, ESA, and our method, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

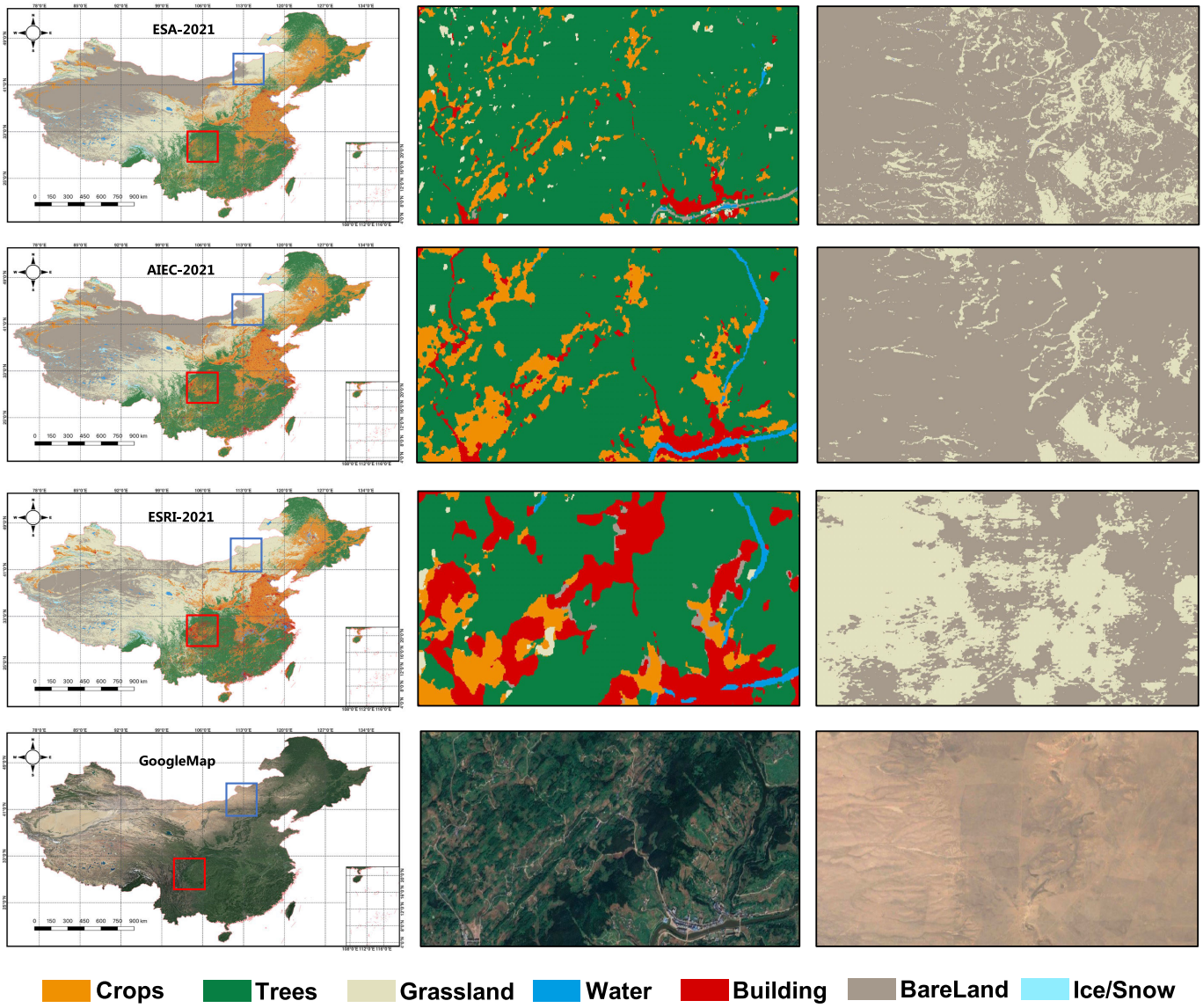


Fig. 9. Visual comparison with other LULC products in China. The left column displays an overall view of China with red and blue boxes marking significant differences. The middle column shows a close-up of the red box at a resolution of 5 m, and the right column displays a close-up of the blue box at a resolution of 5 m. The bottom row displays a Google Map of the corresponding location.

and PLR helps to improve the prediction performance of Dual-Net.

4) The addition of PLR leads to only marginal improvements in overall performance, but the improvement in local details is noticeable. Fig. 7 shows the visual changes brought about by PLR. We observe that narrow roads, small crops, and thin rivers are recognized more accurately.

C. Comparison Between One-Date and Two-Date Inputs

Table IV compares the performance of Dual-Net when provided with either single-date or paired dual-date imagery as input. As demonstrated, using dual-date images leads to substantial gains in performance, with improvements of 5.10 and 2.23 in mean IoU on LCND and CSVD, respectively. These results also indicate that categories associated with vegetation, such as *trees* and *crops*, tend to benefit more significantly from dual-date images, while nonvegetation classes exhibit relatively minor increases. This is likely because vegetation classes often experience more rapid and discernible phenological changes over time compared with nonvegetated classes. By incorporating dual-date data, the model can leverage temporal differences to distinguish these dynamic vegetation categories, whereas single-date data lacks such temporal information.

D. Comparison With Other LULC Products

We compare our product to three existing 10-m-resolution LULC products: FROM10, ESRI, and ESA. FROM10 provided results only for 2017, while ESRI and ESA obtained results for 2020 and 2021. We first reproject each product to the WGS84 Coordinate System (EPSG CODE: 4326), then cut

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

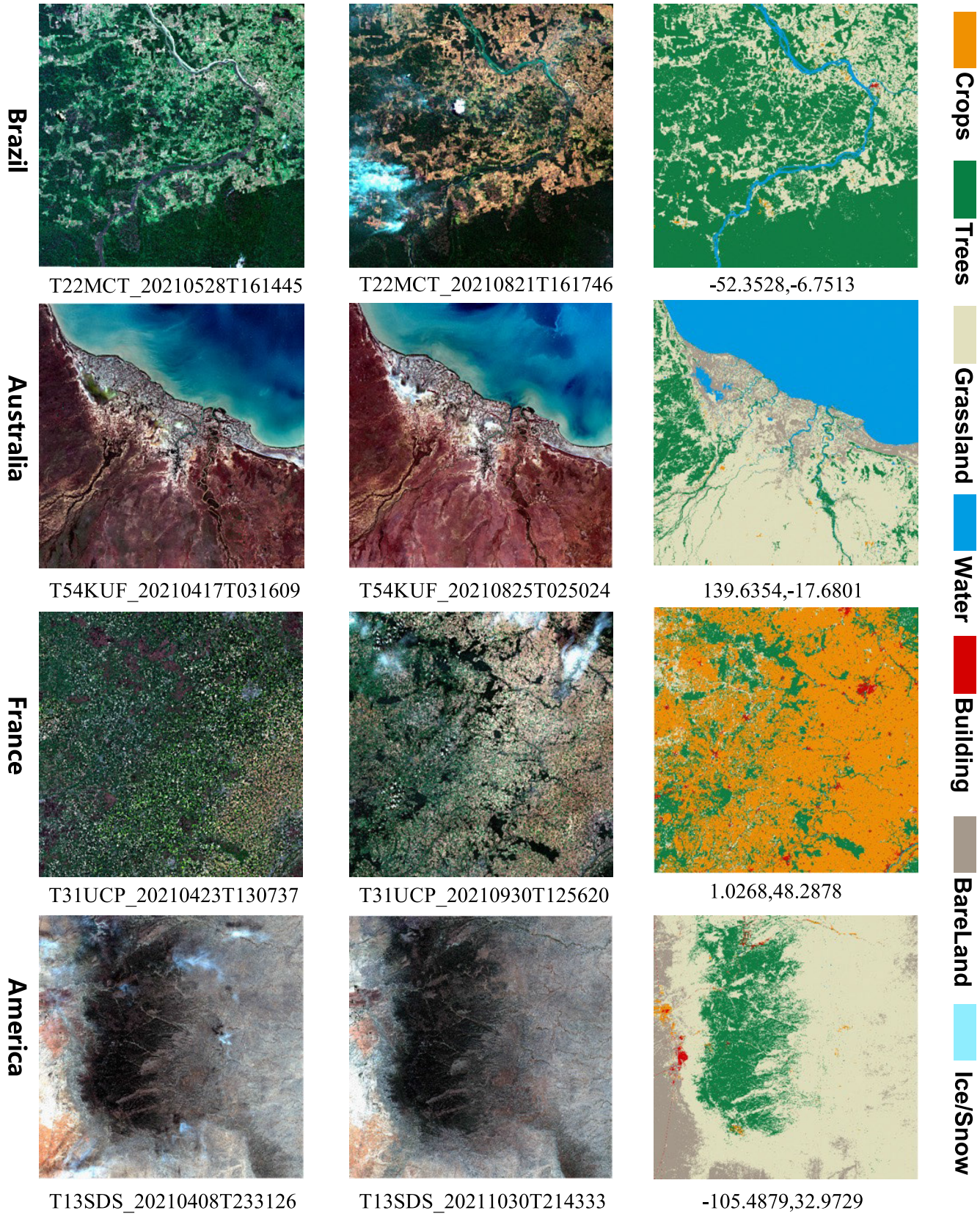


Fig. 10. Visualizations of LULC map predictions for four locations beyond the training data coverage area. The first two columns present Sentinel-2 input image pairs captured on two distinct dates, while the third column exhibits the corresponding predicted LULC maps. Beneath each input image, the Sentinel-2 tile identifier and acquisition date are provided. Under each predicted map, the longitude and latitude coordinates of the visualization extent are displayed.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

out the corresponding positions based on the location of each tile in the validation set. We then remove wetlands, shrubs, and other irrelevant categories and evaluate the performance of the resulting clean, cropped tiles. To ensure a fair comparison, we also generated our own LULC product in 2021, which we called AIEC (AI Earth in China).

The results are presented in Table V, with the performance in each category shown separately. AIEC-2021 refers to the result produced by our released LULC product, while LCND-2018 and CSVD-2021 refer to the metrics calculated directly from the annotation pairs. Since DWD did not offer paired images for two dates, we ignore its evaluation by annotation pairs. Compared with similar class taxonomy products, our AIEC-21 achieved 72.04 mIoU, 68.27 mIoU, and 68.74 mIoU on three validation datasets, significantly outperforming FROM10 and ESA. However, because ESRI has a different definition of crops and bare land, the scores for these categories are lower than those of the other products. When compared with categories without definition divergence, our performances still outperformed ESRI. When using label pairs, our approach achieves higher quantitative results than those produced in the deployment environment. Overall, it is clear that our approach outperforms all other methods, proving its effectiveness.

VII. Visual Comparison

A. Local Comparison

The results presented in Fig. 8 reveal that the land cover classifications produced by ESRI (the second column) exhibit a relatively coarse level of detail, with smoothed boundaries and an inability to detect minute objects or narrow roads, owing to its downsampling model structure. In contrast, our approach, which also employs a CNN-based methodology, produces results that capture more intricate details and accurately fit object boundaries. Furthermore, when compared with the pixel-based ESA approach (the third column), which excels in capturing detailed objects, such as crooked paths in farmland, isolated meadows in mountainous woods, and narrow tree cover between crop lands, the ESA approach struggles with semantic errors, such as misclassifying an entire road as bare land. In contrast, our approach generally yields coarser results than ESA but performs well in semantic recognition. Specifically, we are able to accurately segment large areas of bare land and discern narrow roads. This is attributed to our Dual-Net model, which leverages contextual clues from neighboring pixels and integrates date and location information to jointly classify land cover types.

B. Global Comparison

The visual comparison of three LULC products is shown in Fig. 9. It can be seen that the land cover distribution is fairly similar among the products, with the western region being mostly barren, the south having a significant amount of forests, and the central region is mainly composed of farmland. However, there are two areas that stand out as different: 1) the distinction between crops and trees in Sichuan (highlighted in the red box in Fig. 9) and 2) the classification of grassland and bare land in Inner Mongolia (highlighted in the blue box).

We use Google Maps to compare the details of three products at a resolution of 5 m for the Sichuan region and Inner Mongolia region. In the Sichuan region, we find that ESRI incorrectly classifies some crops as trees and that ESA misidentifies some trees as buildings. Our results can accurately distinguish between trees and crop areas, including the segmentation of smaller features, such as roads and rivers. In Inner Mongolia, which is located at the border between grassland and bare land, the two types of land are mainly distinguished by the percentage of grass cover or the vegetation index. FROM10 defines bare land as having less than 15% grass cover, while ESA defines it as less than 10%. ESRI has a lower percentage of grass cover for this area.

Fine-grained classification is a challenge when it comes to distinguishing between grassland and bare land. The results are heavily influenced by the grassland's vegetation index, which is higher in the summer and weaker in the spring and fall, making it easier to classify as bare land. Since our data products primarily use data from the four months with the lowest cloud cover (April, May, September, and October), we predict more bare land in this transition zone. When comparing the images in the right column and using Google Maps as a reference, it is difficult to visually distinguish between grassland and bare land. Our method extracts areas with the most obvious vegetation index as grassland, while ESA extracts relatively more grass and ESRI extracts the most. However, the distribution between the images and predictions is not quite in agreement. Overall, compared with ESRI, which also uses deep learning methods, we are able to find smaller targets and achieve comparable refinement as the pixel-based method used by ESA.

C. Visual Validation of Generalization

To validate our model's generalization, we selected four regions across the globe for LULC mapping, located in Brazil, France, the United States, and Australia, respectively, and each geographically remote from our train district. We employ Dual-Net without PAAB block to generate LULC maps for these locations, which are shown in Fig. 10. Based on the visualized model predictions, we can draw two preliminary conclusions: 1) our model generated coherent predictions for these regions, indicating its ability to generalize to new geographic areas. 2) Regions from Brazil, France, and the United States, contained cloud-obscured areas in one or both images. However, the model predictions in cloudy and noncloudy regions were consistent and spatially smooth, highlighting the robustness of our approach to limited cloud contamination. These results provide evidence of our model's generalization capabilities and its applicability to other regions of the world.

VIII. Conclusion

In this study, we have created yearly LULC products for the years 2020–2022, which can be used in fields related to LULC. We also provide a real-time online prediction interface, which allows users to get results in real time on our platform.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

Technically, we created training datasets from preexisting data products, enhanced model performance by combining them with multimodal information (date and location), and then used low-level constraints and pseudolabeling to achieve finer segmentation. Our Dual-Net is flexible and can allow either single or two inputs. In conclusion, we offer a low-cost, higher accurate solution for the LULC task that is easy to adapt to other regions. Additionally, we provide a carefully annotated validation dataset containing 733 finely labeled patches with a size of 1024×1024, which is currently the largest validation dataset for China. This dataset can be used as a reference by other researchers and encourages the development of land cover classification.

While the proposed methodology demonstrates promise, there exist certain limitations that preclude optimal performance. The primary constraints on the approach can be distilled into four core aspects. 1) The training data for our method is built upon the prediction data of existing approaches, which constrains the predicted land cover categories to the categories available in existing products; 2) given the long-tailed distribution of land cover classes in the study area, there are less shrub-scrub and wetland classes when employing uniform or random sampling methods for training patch. Consequently, we exclude these two classes from our classification taxonomy. If we elect to include these classes, we should selectively choose training patches from regions that exhibit a greater abundance of shrub-scrub and wetland; 3) Relative to a single-neural network model, the dual-stream architecture of Dual-Net is computationally more complex during inference due to the additional parameters; and 4) differentiating between land covers with similar spectral signatures across two images can be challenging. In practice, we mitigate this case by enlarging the date gap between input images as much as possible because this can introduce more distinctive appearances.

REFERENCES

[1] R. Thackway, L. Lymburner, and J. P. Guerschman, "Dynamic land cover information: Bridging the gap between remote sensing and natural resource management," *Ecol. Soc.*, vol. 18, no. 1, p. 2, 2013.

[2] C. Xiuwan, "Using remote sensing and GIS to analyse land cover change and its impacts on regional sustainable development," *Int. J. Remote Sens.*, vol. 23, no. 1, pp. 107–124, Jan. 2002.

[3] M. L. Cadenasso, S. T. A. Pickett, and K. Schwarz, "Spatial heterogeneity in urban ecosystems: Reconceptualizing land cover and a framework for classification," *Frontiers Ecol. Environ.*, vol. 5, no. 2, pp. 80–88, Mar. 2007.

[4] P. H. Verburg, K. Neumann, and L. Nol, "Challenges in using land use and land cover data for global change studies," *Global Change Biol.*, vol. 17, no. 2, pp. 974–989, Feb. 2011.

[5] S. Fritz, L. See, and F. Rembold, "Comparison of global and regional land cover maps with statistical information for the agricultural domain in Africa," *Int. J. Remote Sens.*, vol. 31, no. 9, pp. 2237–2256, May 2010.

[6] J. M. B. Carreiras, Y. E. Shimabukuro, and J. M. C. Pereira, "Fraction images derived from SPOT-4 VEGETATION data to assess land-cover change over the State of Mato Grosso, Brazil," *Int. J. Remote Sens.*, vol. 23, no. 23, pp. 4979–4983, Jan. 2002.

[7] P. Mayaux, E. Bartholomé, S. Fritz, and A. Belward, "A new land-cover map of Africa for the year 2000," *J. Biogeogr.*, vol. 31, no. 6, pp. 861–877, May 2004.

[8] M. A. Friedl et al., "Global land cover mapping from MODIS: Algorithms and early results," *Remote Sens. Environ.*, vol. 83, nos. 1–2, pp. 287–302, Nov. 2002.

[9] C. Giri, Z. Zhu, and B. Reed, "A comparative analysis of the global land cover 2000 and MODIS land cover data sets," *Remote Sens. Environ.*, vol. 94, no. 1, pp. 123–132, Jan. 2005.

[10] S. Sterckx et al., "The PROBA-V mission: Image processing and calibration," *Int. J. Remote Sens.*, vol. 35, no. 7, pp. 2565–2588, Apr. 2014.

[11] Y. E. Shimabukuro et al., "Discriminating land use and land cover classes in Brazil based on the annual PROBA-V 100 m time series," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3409–3420, 2020.

[12] J. R. Irons, J. L. Dwyer, and J. A. Barsi, "The next Landsat satellite: The Landsat data continuity mission," *Remote Sens. Environ.*, vol. 122, pp. 11–21, Jul. 2012.

[13] M. C. Hansen and T. R. Loveland, "A review of large area monitoring of land cover change using Landsat data," *Remote Sens. Environ.*, vol. 122, pp. 66–74, Jul. 2012.

[14] M. Main-Knorn, B. Pflug, J. Louis, V. Debaecker, U. Müller-Wilm, and F. Gascon, "Sen2cor for Sentinel-2," *Proc. SPIE*, vol. 10427, pp. 37–48, Oct. 2017.

[15] D. Phiri, M. Simwanda, S. Salekin, V. Nyirenda, Y. Murayama, and M. Ranagalage, "Sentinel-2 data for land cover/use mapping: A review," *Remote Sens.*, vol. 12, no. 14, p. 2291, Jul. 2020.

[16] L. Liu, X. Zhang, Y. Gao, X. Chen, X. Shuai, and J. Mi, "Finer-resolution mapping of global land cover: Recent developments, consistency analysis, and prospects," *J. Remote Sens.*, vol. 2021, pp. 1–38, Jan. 2021.

[17] E. Bartholome et al., "GLC 2000: Global land cover mapping for the year 2000," Project Status, Joint Res. Centre, Inst. Environ. Sustainability, Eur. Commission, Ispra, Italy, Tech. Rep., Nov. 2002.

[18] R. Latifovic, Z.-L. Zhu, J. Cihlar, C. Giri, and I. Olthof, "Land cover mapping of North and Central America—Global land cover 2000," *Remote Sens. Environ.*, vol. 89, no. 1, pp. 116–127, 2004.

[19] M. A. Friedl et al., "MODIS collection 5 global land cover: Algorithm refinements and characterization of new datasets," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 168–182, Jan. 2010.

[20] O. Arino, J. J. R. Perez, V. Kalogirou, S. Bontemps, P. Defourny, and E. Van Bogaert, "Global land cover map for 2009 (GlobCover 2009)," Eur. Space Agency, Université Catholique de Louvain, Frascati, Italy, Tech. Rep., 2012.

[21] J. Chen et al., "Global land cover mapping at 30m resolution: A POK-based operational approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 103, pp. 7–27, May 2015.

[22] P. Gong et al., "Finer resolution observation and monitoring of global land cover: First mapping results with Landsat TM and ETM+ data," *Int. J. Remote Sens.*, vol. 34, no. 7, pp. 2607–2654, Apr. 2013.

[23] L. Yu, J. Wang, and P. Gong, "Improving 30 m global land-cover map FROM-GLC with time series MODIS and auxiliary data sets: A segmentation-based approach," *Int. J. Remote Sens.*, vol. 34, no. 16, pp. 5851–5867, Aug. 2013.

[24] X. Zhang, L. Liu, X. Chen, Y. Gao, S. Xie, and J. Mi, "GLC_FCS30: Global land-cover product with fine classification system at 30 m using time-series Landsat imagery," *Earth Syst. Sci. Data*, vol. 13, no. 6, pp. 2753–2776, Jun. 2021.

[25] P. Gong et al., "Stable classification with limited sample: Transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017," *Sci. Bull.*, vol. 64, no. 6, pp. 370–373, Mar. 2019.

[26] D. Zanaga et al., "ESA WorldCover 10 m 2021 V200," Eur. Space Agency, Paris, France, Tech. Rep., 2022, doi: 10.5281/zenodo.7254221.

[27] K. Karra, C. Kontgis, Z. Statman-Weil, J. C. Mazzariello, M. Mathis, and S. P. Brumby, "Global land use/land cover with Sentinel 2 and deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 4704–4707.

[28] C. F. Brown et al., "Dynamic world, near real-time global 10 m land use land cover mapping," *Sci. Data*, vol. 9, no. 1, pp. 1–17, Jun. 2022.

[29] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," 2017, *arXiv:1709.00029*.

[30] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[31] J. Xia, N. Yokoya, B. Adriano, and C. Broni-Bediako, "OpenEarthMap: A benchmark dataset for global high-resolution land cover mapping," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6243–6253.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.

[32] P. Kaiser, J. D. Wegner, A. Lucchi, M. Jaggi, T. Hofmann, and K. Schindler, "Learning aerial image segmentation from online maps," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6054–6068, Nov. 2017.

[33] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 28–37.

[34] Y. Long et al., "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-AID," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021.

[35] J. D. Paola and R. A. Schowengerdt, "A detailed comparison of backpropagation neural network and maximum-likelihood classifiers for urban land use classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 4, pp. 981–996, Jul. 1995.

[36] D. L. Civco, "Artificial neural networks for land-cover classification and mapping," *Int. J. Geograph. Inf. Syst.*, vol. 7, no. 2, pp. 173–186, Mar. 1993.

[37] M. Pal and P. M. Mather, "An assessment of the effectiveness of decision tree methods for land cover classification," *Remote Sens. Environ.*, vol. 86, no. 4, pp. 554–565, Aug. 2003.

[38] S. Talukdar et al., "Land-use land-cover classification by machine learning classifiers for satellite observations—A review," *Remote Sens.*, vol. 12, no. 7, p. 1135, Apr. 2020.

[39] T. Kavzoglu and I. Colkesen, "A kernel functions analysis for support vector machines for land cover classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 11, no. 5, pp. 352–359, Oct. 2009.

[40] K. O'Shea and R. Nash, "An introduction to convolutional neural networks," 2015, *arXiv:1511.08458*.

[41] W. Li, R. Dong, H. Fu, J. Wang, L. Yu, and P. Gong, "Integrating Google Earth imagery with Landsat data to improve 30-m resolution land cover mapping," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111563.

[42] D. Verma and A. Jana, "LULC classification methodology based on simple convolutional neural network to map complex urban forms at finer scale: Evidence from Mumbai," 2019, *arXiv:1909.09774*.

[43] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Munich, Germany: Springer, Oct. 2015, pp. 234–241.

[44] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[45] Z. Li, H. Zhang, F. Lu, R. Xue, G. Yang, and L. Zhang, "Breaking the resolution barrier: A low-to-high network for large-scale high-resolution land-cover mapping using low-resolution labels," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 244–267, Oct. 2022.

[46] H. Alemohammad and K. Booth, "LandCoverNet: A global benchmark land cover classification training dataset," 2020, *arXiv:2012.03111*.

[47] X. Xu, C. Lu, X. Shi, and Y. Ding, "Large-scale topography of China: A factor for the seasonal progression of the meiyu rainband?" *J. Geophys. Res.*, vol. 115, no. D2, p. D02110, 2010.

[48] A. Di Gregorio, *Land Cover Classification System: Classification Concepts and User Manual: LCCS*, vol. 2. Rome, Italy: Food & Agriculture Organization, 2005.

[49] Q. Zhao, S. Lyu, Y. Li, Y. Ma, and L. Chen, "MGML: Multigranularity multilevel feature ensemble network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 5, pp. 2308–2322, May 2023.

[50] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606216.

[51] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3500–3509.

[52] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.

[53] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[54] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1510–1519.

[55] K. Jiang, Z. Wang, P. Yi, and J. Jiang, "A progressively enhanced network for video satellite imagery superresolution," *IEEE Signal Process. Lett.*, vol. 25, no. 11, pp. 1630–1634, Nov. 2018.

[56] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[57] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[58] T. Ojala, M. Pietikäinen, and D. Harwood, "A comparative study of texture measures with classification based on featured distributions," *Pattern Recognit.*, vol. 29, no. 1, pp. 51–59, Jan. 1996.

[59] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, May 1979.

[60] S. K. McFeeters, "The use of the normalized difference water index (NDWI) in the delineation of open water features," *Int. J. Remote Sens.*, vol. 17, no. 7, pp. 1425–1432, May 1996.

[61] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth engine: Planetary-scale geospatial analysis for everyone," *Remote Sens. Environ.*, vol. 202, pp. 18–27, Dec. 2017.

[62] X. Zhang, L. Liu, X. Chen, S. Xie, and Y. Gao, "Fine land-cover mapping in China using Landsat datacube and an operational SPECLib-based approach," *Remote Sens.*, vol. 11, no. 9, p. 1056, May 2019.

[63] S. Manzanarez, V. Manian, and M. Santos, "Land use land cover labeling of GLOBE images using a deep learning fusion model," *Sensors*, vol. 22, no. 18, p. 6895, Sep. 2022.

[64] A. Buslaev, V. I. Iglovikov, E. Khvedchenya, A. Parinov, M. Druzhinin, and A. A. Kalinin, "Albumentations: Fast and flexible image augmentations," *Information*, vol. 11, no. 2, p. 125, Feb. 2020.

[65] MMS Contributors. (2020). *MMSegmentation: Openmmlab Semantic Segmentation Toolbox and Benchmark*. https://github.com/open-mmlab/mmsegmentation

[66] D. Zanaga et al., "ESA WorldCover 10 m 2020 v100 (version v100)[data set]," Zenodo, 2021.



Shang Liu received the M.Sc. degree from the University Liaoning Technical, Fuxin, China, in 2016.

He is currently a Senior Algorithm Scientist with the DAMO Academy, Alibaba Group, Beijing, China. His research interests include computer vision and remote sensing, with an emphasis on semantic segmentation, unsupervised learning, and self-supervised learning.



Huadong Wang received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2017.

He was with Samsung Research China-Beijing, Beijing, from 2017 to 2021. He is currently a Post-Doctoral Researcher with the Department of Computer Science and Technology, Tsinghua University, Beijing. His research interests include machine learning, natural language processing, and foundation model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.



Yuan Hu received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2020.

She is currently a Senior Algorithm Scientist with the DAMO Academy, Alibaba Group, Beijing. Her research interests include deep learning, computer vision, and remote sensing.



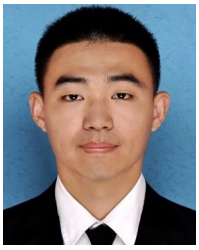
Mengting Zhang received the M.Sc. degree in meteorology from the Chinese Academy of Agricultural Sciences, Beijing, China, in 2017.

She is currently working with AI Earth, DAMO Academy, Alibaba Group, Beijing, where she is involved in remote sensing and meteorological application-related works. Her research interests include deep learning, remote sensing, and agricultural systems.



Yixuan Zhu received the M.S. degree from Wuhan University, Wuhan, China, in 2022.

He is currently with the Alibaba Group, Beijing, China. His research interests include artificial intelligence engineering, deep learning engineering, distributed computing, and cloud computing.



Zhibin Wang received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2015.

He is currently the Staff Algorithm Scientist with the DAMO Academy, Alibaba Group, Hangzhou, China. His research interests include deep learning, computer vision, and remote sensing.



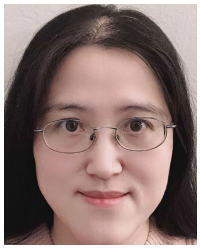
Dongyang Li received the M.S. degree from Tianjin University, Tianjin, China, in 2017.

He is currently a Senior Algorithm Scientist with the DAMO Academy, Alibaba Group, Beijing, China. His research interests include deep learning, computer vision, and video coding.



Mingyang Yang received the B.Sc. degree from Nanjing University, Nanjing, China, in 2012, and the M.Eng. degree from Shanghai Jiao Tong University, Shanghai, China, in 2015.

He is currently a Senior Software Engineer with the DAMO Academy, Alibaba Group, Beijing, China. His research interests include artificial intelligence engineering, deep learning engineering, distributed computing, and cloud computing.



Fan Wang received the B.S. and M.S. degrees from the Department of Automation, Tsinghua University, Beijing, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Department of Electrical Engineering, Stanford University, Stanford, CA, USA, in 2015.

She is currently a Senior Staff Algorithm Engineer with the Alibaba Group, Beijing. Her research interests include object tracking and recognition, image segmentation, 3-D vision, and multisensor fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:25:37 UTC from IEEE Xplore. Restrictions apply.